GraM-Diff: A Unified Graph-Mamba Diffusion Framework for EEG-Based Alzheimer's Disease Data Generation and Diagnosis

M. Tanveer Ayush Singh Rana Sanskriti Jain Arnav Kumar Aryaman Tiwari Abdur Rahaman Abdul Quadir M. Sajid

OPTIMAL Research Lab, Indian Institute of Technology Indore, Madhya Pradesh, India

MICCAI 2026 EMERGE 2026
Archival Track
Submission #25

Introduction

Alzheimer's Disease (AD) is a leading cause of dementia, requiring early and scalable diagnostic strategies. While Electroencephalography (EEG) offers a non-invasive, cost-effective modality with high temporal resolution for monitoring brain activity, its clinical potential is currently bottlenecked by data limitations.

Key idea: **GraM-Diff** is a unified, classifier-guided Graph-Mamba diffusion framework that resolves these limitations through reliable, high-fidelity synthetic EEG generation.

Background & Motivation

- **Data Scarcity & Imbalance:** Deep learning models for EEG are fundamentally constrained by the limited scale and extreme class imbalance of clinically available datasets.
- **Failure of Current Generators:** Existing generative models fail to explicitly capture the *non-Euclidean connectivity* of the brain and *long-range temporal dependencies* essential for EEG signals.

Proposed Methodology

GraM-Diff introduces four key innovations to synthesize topologically and temporally accurate EEG data:

1. **Topology-Aware Spatial Encoding:** GCNs explicitly capture the non-Euclidean spatial relationships among electrodes.
2. **Bidirectional Mamba Bottleneck:** Efficient linear-complexity ($\mathcal{O}(L)$) State Space Models capture long-range temporal dependencies, avoiding quadratic Transformer costs.
3. **Latent-Space Classifier Guidance:** Conditions a single diffusion model on the target disease state within a shared latent space.
4. **Physiological Decomposition:** End-to-end separation of slow-varying morphological trends and fast oscillatory rhythms (seasonality).

Visual Architecture of GraM-Diff

End-to-End Generative Pipeline: GraM-Diff integrates spatial graph learning with temporal sequence modeling to generate clinically meaningful EEG representations. The GCN captures inter-electrode relationships, while the Bidirectional Mamba bottleneck models long-range temporal dependencies. The decoder reconstructs the EEG signal while separating its morphological and oscillatory components.

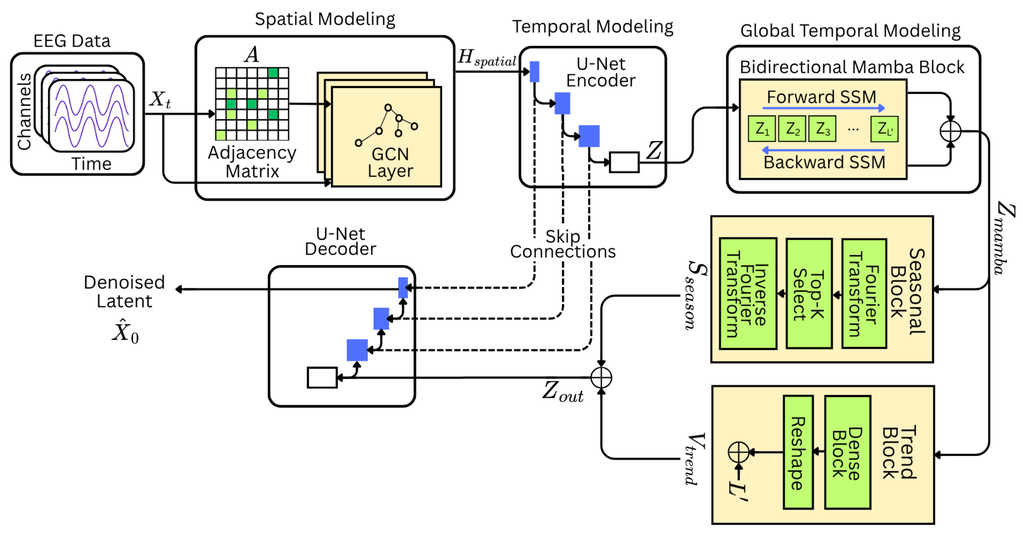


Figure 1. The noisy EEG signal is processed through a spatial GCN, compressed via a U-Net Encoder, and passed into a linear-complexity Bidirectional Mamba bottleneck. The decoder reconstructs and decomposes the signal.

Mathematical Formulation

Diffusion Process:

The forward noising process utilizes a Cosine Schedule:

$$q(\mathbf{X}_t \mid \mathbf{X}_0) = \mathcal{N}\left(\mathbf{X}_t; \sqrt{\bar{\alpha}_t}\,\mathbf{X}_0, (1-\bar{\alpha}_t)\mathbf{I}\right)$$

Topology-Aware Spatial Encoding:

Using physical 3D coordinates from the 10-20 system, spatial topology is computed via:

$$\mathbf{H}_{\text{spatial}} = \text{LayerNorm}\left(\sigma\left(\hat{\mathbf{A}}\mathbf{X}_t\mathbf{W}_{\text{gnn}}\right)\right)$$

Dual-Domain Optimization:

To preserve temporal dynamics and explicit oscillatory brain rhythms, we optimize via time and frequency domains:

$$\mathcal{L} = \underbrace{\|\mathbf{X}_0 - \hat{\mathbf{X}}_0\|_1}_{\text{Time-Domain}} + \lambda \underbrace{\sum_{p\in\{\text{Re},\text{Im}\}} \left\| p(\mathcal{F}(\mathbf{X}_0)) - p(\mathcal{F}(\hat{\mathbf{X}}_0)) \right\|_1}_{\text{Frequency-Domain Spectral Penalty}}$$

Experimental Results

1. Comparison with Generative Peer Models (ADSZ Dataset)

Comparison of downstream classification accuracy when trained on synthesized data across various state-of-the-art generators.

Model	Accuracy	Precision	F1	AUROC
TimeVAE	64.25%	68.11%	69.74%	65.29%
TimeGAN	69.81%	70.28%	71.93%	72.29%
Diffusion-TS	87.41%	89.25%	88.65%	90.83%
GraM-Diff (Ours)	**97.18%**	**97.20%**	**96.28%**	**97.82%**

2. Generative Fidelity Metrics

Evaluating how closely generated data aligns with actual clinical EEG distributions via deep feature distance (Context-FID) and Correlation Scores.

Metric	ADSZ		APAVA	
	Ours	Diff-TS	Ours	Diff-TS
Context-FID ↓	**0.109**	0.585	**0.286**	0.426
Corr. Score ↓	**0.275**	0.492	**0.402**	0.829

Note: GraM-Diff severely minimizes Context-FID and achieves superior Correlation Scores, confirming it preserves complex temporal and spatial inter-channel dependencies.

Conclusion & Computational Efficiency

- **High Efficiency:** Replaces $\mathcal{O}(L^2)$ Transformer bottlenecks with linear scaling. Generates 1000 synthetic samples in ~6 minutes.
- **Manifold Smoothing:** TSTR (Synthetic Train) > TRTR (Real Train) phenomenon demonstrates the model densely populates the data manifold, resolving the small-data challenge in Alzheimer's diagnosis.

References

[1] Ho, J., Jain, A., Abbeel, P.: Denoising diffusion probabilistic models. Advances in Neural Information Processing Systems (2020)
[2] Klepl, D., He, F., Wu, M., De Marco, M., Blackburn, D.: EEG-based graph neural network classification of Alzheimer's disease. IEEE/ACM Transactions on Computational Biology and Bioinformatics (2022)
[3] Gui, J., et al.: EEG-Mamba: A Bidirectional State Space Model for EEG Signal Classification. arXiv preprint (2024)

Contact: M. Sajid
phd2101241003@iiti.ac.in | mohammad.sajid@ntu.edu.sg

Code
https://github.com/mtanveer1/GraM-Diff

Access the Paper: